IND320-1 26H Data til beslutning

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GitHub · Streamlit app · Screencast

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Execution: The public NVE cells run without credentials. ENTSO-E and MongoDb Atlas database cells are applied with e-token and cluster's password.

0. General

Setup & Admin

- ☐ **Git Workflow:** Work in a dedicated feature branch. Merge into `main` only after peer review and teacher feedback.
- ☐ **Credentials:** Set up security tokens for ENTSO-E and prepare MongoDB credentials (store in Streamlit Cloud secrets, never hardcode).
- ☐ **Databases & Spark:** Verify local Cassandra and Spark installation; test Spark–Cassandra and Python–MongoDB connections.

1. Jupyter Notebook (Local Core)

- ☐ **Magasinstatistikk API:** Replace old CSV data source with direct API calls; recreate Part 1 plots.
- ☐ **ENTSO-E API:** Fetch cross-border transfer data (Norway, last 2 years) and save raw data into Cassandra via Spark.
- ☐ **Spark Processing:** Extract filtered columns from Cassandra and build two visualizations:
- Pie chart showing annual import vs. export for a selected connection.
- Line plot showing the first month’s flow per connection and direction.
- ☐ **Database Transfer:** Export curated Spark data into MongoDB.
- ☐ **Documentation:** Add code comments, run all cells to show output before exporting to PDF, and include:

- Public GitHub & Streamlit app links.
- AI tools usage summary.
- 300–500 word work log (Jupyter + Streamlit process).
- [ ] Push `.ipynb` file to the GitHub repository.

2. Streamlit App (Community Cloud)

- [ ] **Page Layout Updates:**
- Shift existing Page 4 content to Page 5 if needed.
- Logical UI restructuring across pages 2–3 is permitted.
- [ ] **Magasinstatistikk Integration:**
- Connect directly to the API (no CSV).
- Use selection slider to set data download interval.
- Add `st.radio` for area type (EL, NO, VASS) to dynamically update plots and `LineChartColumn()`.
- [ ] **MongoDB Integration:**
- Connect securely using Streamlit Secrets.
- Create a two-column layout (`st.columns`):
- **Left Column:** `st.radio` (connection) + year picker → Annual Import/Export Pie Chart.
- **Right Column:** `st.pills` (month selection) → Multi-month Line Plot.
- Below columns: Add `st.expander` with data source documentation.

3. Screencast & Submission

- [ ] Record a 5-minute video demonstrating the live app and explaining key code segments.
- [ ] Submission includes PDF export of Notebook, `.ipynb` in repo, and Screencast link (via Panopto/GitHub link in comments/notebook).

Two APIs, why?

API	Data provided	Main question
NVE Magasinstatistikk	Reservoir filling levels, storage capacity and stored energy in Norway	How full are Norway's reservoirs?
ENTSO-E	Electricity transfers between Norway and neighbouring countries	How much electricity enters and leaves Norway?

1. NVE: replace the reservoir CSV

Fetch reservoir statistics directly from the API instead of reading `reservoirs.csv`, then recreate the existing reservoir plots.

Flow: NVE API → reservoir analysis → Streamlit

2. ENTSO-E: add electricity-transfer analysis

Fetch cross-border transfers for the last two years, process and store the observations, and create the required pie and line charts.

Flow: ENTSO-E → Spark → Cassandra → Spark → MongoDB → Streamlit

1. Setup

Use **Python 3.11**, **Java 17**, and Cassandra at `127.0.0.1:9042`. Install once, restart the kernel, and continue below. The accompanying Docker Compose file starts a local Cassandra server.

In [1]:
```
%pip install -r requirement.txt
```

```
Requirement already satisfied: pyspark==3.5.9 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 2)) (3.5.9)
Requirement already satisfied: pandas==2.3.3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 3)) (2.3.3)
Requirement already satisfied: numpy==1.26.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 4)) (1.26.4)
Requirement already satisfied: matplotlib==3.11.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 5)) (3.11.2)
Requirement already satisfied: requests==2.34.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 6)) (2.34.2)
Requirement already satisfied: pymongo==4.18.3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 7)) (4.18.3)
Requirement already satisfied: cassandra-driver==3.30.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 8)) (3.30.1)
Requirement already satisfied: ipykernel==7.4.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 9)) (7.4.0)
Requirement already satisfied: jupyterlab==4.6.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
-r requirement.txt (line 10)) (4.6.4)
Requirement already satisfied: py4j<0.10.9.10,≥0.10.9.7 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
pyspark==3.5.9→-r requirement.txt (line 2)) (0.10.9.9)
Requirement already satisfied: python-dateutil≥2.8.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
pandas==2.3.3→-r requirement.txt (line 3)) (2.9.0.post0)
Requirement already satisfied: pytz≥2020.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
pandas==2.3.3→-r requirement.txt (line 3)) (2026.5)
Requirement already satisfied: tzdata≥2022.7 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
pandas==2.3.3→-r requirement.txt (line 3)) (2026.5)
Requirement already satisfied: contourpy≥1.0.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (1.3.3)
Requirement already satisfied: cycler≥0.10 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (0.12.1)
Requirement already satisfied: fonttools≥4.28.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (4.66.1)
Requirement already satisfied: kiwisolver≥1.3.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (1.5.1)
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Requirement already satisfied: packaging≥20.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (26.3)
Requirement already satisfied: pillow≥9 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (12.3.0)
Requirement already satisfied: pyparsing≥3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
matplotlib==3.11.2→-r requirement.txt (line 5)) (3.3.3)
Requirement already satisfied: charset_normalizer<4,≥2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
requests==2.34.2→-r requirement.txt (line 6)) (3.5.2)
Requirement already satisfied: idna<4,≥2.5 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
requests==2.34.2→-r requirement.txt (line 6)) (3.20)
Requirement already satisfied: urllib3<3,≥1.26 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
requests==2.34.2→-r requirement.txt (line 6)) (2.8.0)
Requirement already satisfied: certifi≥2023.5.7 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
requests==2.34.2→-r requirement.txt (line 6)) (2026.7.22)
Requirement already satisfied: dnspython<3.0.0,≥2.7.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
pymongo==4.18.3→-r requirement.txt (line 7)) (2.9.0)
Requirement already satisfied: geomet≥1.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
cassandra-driver==3.30.1→-r requirement.txt (line 8)) (1.1.0)
Requirement already satisfied: Deprecated≥1.3.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
cassandra-driver==3.30.1→-r requirement.txt (line 8)) (1.3.1)
Requirement already satisfied: comm≥0.1.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (0.2.3)
Requirement already satisfied: debugpy≥1.6.5 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (1.8.22)
Requirement already satisfied: ipython≥7.23.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (9.17.1)
Requirement already satisfied: jupyter-client≥8.9.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (8.10.0)
Requirement already satisfied: jupyter-core≠6.0.*,≥5.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (5.9.1)
Requirement already satisfied: matplotlib-inline≥0.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (0.2.2)
Requirement already satisfied: nest-asyncio2≥1.7.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (1.7.4)
Requirement already satisfied: pyzmq≥25 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (27.2.0)
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Requirement already satisfied: tornado≥6.5.7 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (6.5.10)
Requirement already satisfied: traitlets≥5.4.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipykernel==7.4.0→-r requirement.txt (line 9)) (5.16.1)
Requirement already satisfied: async-lru≥1.0.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (2.3.0)
Requirement already satisfied: httpx<1,≥0.25.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (0.28.1)
Requirement already satisfied: jinja2≥3.0.3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (3.1.6)
Requirement already satisfied: jupyter-builder≥1.0.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (1.2.3)
Requirement already satisfied: jupyter-lsp≥2.0.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (2.3.1)
Requirement already satisfied: jupyter-server<3,≥2.19.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (2.21.1)
Requirement already satisfied: jupyterlab-server<3,≥2.28.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (2.28.1)
Requirement already satisfied: notebook-shim≥0.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (0.2.4)
Requirement already satisfied: typing-extensions≥4.4.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab==4.6.4→-r requirement.txt (line 10)) (4.16.0)
Requirement already satisfied: anyio in /home/tyyniuni/miniconda3/envs/nve-
spark/lib/python3.11/site-packages (from httpx<1,≥0.25.0
→jupyterlab==4.6.4→-r requirement.txt (line 10)) (4.15.1)
Requirement already satisfied: httpcore==1.* in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
httpx<1,≥0.25.0→jupyterlab==4.6.4→-r requirement.txt (line 10)) (1.0.9)
Requirement already satisfied: h11≥0.16 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
httpcore==1.*→httpx<1,≥0.25.0→jupyterlab==4.6.4→-r requirement.txt (line
10)) (0.16.0)
Requirement already satisfied: argon2-cffi≥21.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(25.1.0)
Requirement already satisfied: jupyter-events≥0.11.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(0.12.1)
Requirement already satisfied: jupyter-server-terminals≥0.4.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
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(0.5.4)
Requirement already satisfied: nbconvert≥6.4.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(7.17.2)
Requirement already satisfied: nbformat≥5.3.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(5.11.1)
Requirement already satisfied: overrides≥5.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(7.7.0)
Requirement already satisfied: prometheus-client≥0.9 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(0.26.0)
Requirement already satisfied: send2trash≥1.8.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(2.1.0)
Requirement already satisfied: terminado≥0.8.3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(0.18.1)
Requirement already satisfied: websocket-client≥1.7 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(1.9.2)
Requirement already satisfied: babel≥2.10 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r requirement.txt (line
10)) (2.18.0)
Requirement already satisfied: json5≥0.9.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r requirement.txt (line
10)) (0.17.3)
Requirement already satisfied: jsonschema≥4.18.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r requirement.txt (line
10)) (4.26.0)
Requirement already satisfied: argon2-cffi-bindings in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
argon2-cffi≥21.1→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (26.1.0)
Requirement already satisfied: wrapt<3,≥1.10 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
Deprecated≥1.3.1→cassandra-driver==3.30.1→-r requirement.txt (line 8))
(2.5.0)
Requirement already satisfied: click in /home/tyyniuni/miniconda3/envs/nve-
spark/lib/python3.11/site-packages (from geomet≥1.1→cassandra-
driver==3.30.1→-r requirement.txt (line 8)) (8.5.0)
Requirement already satisfied: ipython-pygments-lexers≥1.0.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
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ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (1.1.1)
Requirement already satisfied: jedi≥0.18.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (0.20.0)
Requirement already satisfied: pexpect>4.6 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (4.9.0)
Requirement already satisfied: prompt_toolkit<3.1.0,≥3.0.41 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (3.0.53)
Requirement already satisfied: psutil≥7 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (7.2.2)
Requirement already satisfied: pygments≥2.14.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (2.21.0)
Requirement already satisfied: stack_data≥0.6.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9)) (0.6.3)
Requirement already satisfied: wcwidth≥0.1.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
prompt_toolkit<3.1.0,≥3.0.41→ipython≥7.23.1→ipykernel==7.4.0→-r
requirement.txt (line 9)) (0.9.2)
Requirement already satisfied: parso<0.9.0,≥0.8.6 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jedi≥0.18.2→ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line
9)) (0.8.7)
Requirement already satisfied: MarkupSafe≥2.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jinja2≥3.0.3→jupyterlab==4.6.4→-r requirement.txt (line 10)) (3.0.4)
Requirement already satisfied: attrs≥22.2.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema≥4.18.0→jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (26.1.0)
Requirement already satisfied: jsonschema-specifications≥2023.03.6 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema≥4.18.0→jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (2025.9.1)
Requirement already satisfied: referencing≥0.28.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema≥4.18.0→jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (0.37.0)
Requirement already satisfied: rpds-py≥0.25.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema≥4.18.0→jupyterlab-server<3,≥2.28.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (2026.9.1)
Requirement already satisfied: platformdirs≥2.5 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-core≠6.0.*,≥5.1→ipykernel==7.4.0→-r requirement.txt (line 9))
(4.13.0)
Requirement already satisfied: python-json-logger≥2.0.4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-events≥0.11.0→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (4.2.0)
```

```
Requirement already satisfied: pyyaml ⩾ 5.3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-events ⩾ 0.11.0→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (6.0.3)
Requirement already satisfied: rfc3339-validator in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-events ⩾ 0.11.0→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (0.1.4)
Requirement already satisfied: rfc3986-validator ⩾ 0.1.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jupyter-events ⩾ 0.11.0→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (0.1.1)
Requirement already satisfied: fqdn in /home/tyyniuni/miniconda3/envs/nve-
spark/lib/python3.11/site-packages (from jsonschema[format-nongpl] ⩾ 4.18.0
→jupyter-events ⩾ 0.11.0→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (1.6.0)
Requirement already satisfied: isoduration in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema[format-nongpl] ⩾ 4.18.0→jupyter-events ⩾ 0.11.0→jupyter-
server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r requirement.txt (line 10))
(20.11.0)
Requirement already satisfied: jsonpointer>1.13 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema[format-nongpl] ⩾ 4.18.0→jupyter-events ⩾ 0.11.0→jupyter-
server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r requirement.txt (line 10)) (3.2.1)
Requirement already satisfied: rfc3987-syntax ⩾ 1.1.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema[format-nongpl] ⩾ 4.18.0→jupyter-events ⩾ 0.11.0→jupyter-
server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r requirement.txt (line 10)) (1.1.0)
Requirement already satisfied: uri-template in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema[format-nongpl] ⩾ 4.18.0→jupyter-events ⩾ 0.11.0→jupyter-
server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r requirement.txt (line 10)) (1.3.0)
Requirement already satisfied: webcolors ⩾ 24.6.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
jsonschema[format-nongpl] ⩾ 4.18.0→jupyter-events ⩾ 0.11.0→jupyter-
server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r requirement.txt (line 10))
(25.10.0)
Requirement already satisfied: beautifulsoup4 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbconvert ⩾ 6.4.4→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (4.15.0)
Requirement already satisfied: bleach≠5.0.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
bleach[css]≠5.0.0→nbconvert ⩾ 6.4.4→jupyter-server<3, ⩾ 2.19.0
→jupyterlab═4.6.4→-r requirement.txt (line 10)) (6.4.0)
Requirement already satisfied: defusedxml in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbconvert ⩾ 6.4.4→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (0.7.1)
Requirement already satisfied: jupyterlab-pygments in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbconvert ⩾ 6.4.4→jupyter-server<3, ⩾ 2.19.0→jupyterlab═4.6.4→-r
requirement.txt (line 10)) (0.3.0)
```

```
Requirement already satisfied: mistune<4,≥2.0.3 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbconvert≥6.4.4→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (3.3.4)
Requirement already satisfied: nbclient≥0.5.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbconvert≥6.4.4→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (0.11.0)
Requirement already satisfied: pandocfilters≥1.4.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbconvert≥6.4.4→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (1.5.1)
Requirement already satisfied: webencodings in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
bleach≠5.0.0→bleach[css]≠5.0.0→nbconvert≥6.4.4→jupyter-
server<3,≥2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10)) (0.6.1)
Requirement already satisfied: tinycss2≥1.1.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
bleach[css]≠5.0.0→nbconvert≥6.4.4→jupyter-server<3,≥2.19.0
→jupyterlab==4.6.4→-r requirement.txt (line 10)) (1.5.1)
Requirement already satisfied: fastjsonschema≥2.15 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
nbformat≥5.3.0→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (2.22.2)
Requirement already satisfied: ptyprocess≥0.5 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
pexpect>4.6→ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt (line 9))
(0.7.0)
Requirement already satisfied: six≥1.5 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
python-dateutil≥2.8.2→pandas==2.3.3→-r requirement.txt (line 3)) (1.17.0)
Requirement already satisfied: lark≥1.2.2 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
rfc3987-syntax≥1.1.0→jsonschema[format-nongpl]≥4.18.0→jupyter-
events≥0.11.0→jupyter-server<3,≥2.19.0→jupyterlab==4.6.4→-r
requirement.txt (line 10)) (1.3.1)
Requirement already satisfied: executing≥1.2.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
stack_data≥0.6.0→ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt
(line 9)) (2.3.0)
Requirement already satisfied: asttokens≥2.1.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
stack_data≥0.6.0→ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt
(line 9)) (3.0.2)
Requirement already satisfied: pure-eval in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
stack_data≥0.6.0→ipython≥7.23.1→ipykernel==7.4.0→-r requirement.txt
(line 9)) (0.2.4)
Requirement already satisfied: cffi≥1.0.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
argon2-cffi-bindings→argon2-cffi≥21.1→jupyter-server<3,≥2.19.0
→jupyterlab==4.6.4→-r requirement.txt (line 10)) (2.1.1)
Requirement already satisfied: pycparser in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
```

```
cffi ⩾ 1.0.1→argon2-cffi-bindings→argon2-cffi ⩾ 21.1→jupyter-
server<3, ⩾ 2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10)) (3.11)
Requirement already satisfied: soupsieve ⩾ 1.6.1 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
beautifulsoup4→nbconvert ⩾ 6.4.4→jupyter-server<3, ⩾ 2.19.0
→jupyterlab==4.6.4→-r requirement.txt (line 10)) (3.0.2)
Requirement already satisfied: arrow ⩾ 0.15.0 in
/home/tyyniuni/miniconda3/envs/nve-spark/lib/python3.11/site-packages (from
isoduration→jsonschema[format-nongpl] ⩾ 4.18.0→jupyter-events ⩾ 0.11.0
→jupyter-server<3, ⩾ 2.19.0→jupyterlab==4.6.4→-r requirement.txt (line 10))
(1.4.0)
Note: you may need to restart the kernel to use updated packages.
```

2. NVE: fetch reservoir statistics

This replaces `pd.read_csv("reservoirs.csv")`. Keep the three classifications (`NO`, `EL`, `VASS`) in the dataset. They overlap geographically and must not be added together.

In [2]:
```
import requests
import pandas as pd
import matplotlib.pyplot as plt

url =
"https://biapi.nve.no/magasinstatistikk/api/Magasinstatistikk/HentOffentligData"
response = requests.get(url, timeout=60)
response.raise_for_status()
raw = pd.DataFrame(response.json())
print("Fetched reservoir records:", len(raw))
raw.head()
```

```
Fetched reservoir records: 14922
```

Out[2]: Columns 1–8 of 12

	dato_Id	omrType	omrnr	iso_aar	iso_uke	fyllingsgrad	kapasitet_TWh
0	1995-09-03	EL	4	1995	35	0.944721	21.079208
1	2020-11-15	EL	1	2020	46	0.974361	6.003264
2	2011-08-28	EL	4	2011	34	0.786499	21.079208
3	1998-07-05	EL	3	1998	27	0.793969	8.920932
4	2011-03-27	EL	4	2011	12	0.300820	21.079208

Columns 9–12 of 12 · first column repeated

	fylling_TWh	neste_Publiseringsdato	fyllingsgrad_forrige_uke	endring_fyllingsgrad
0	19.913979	0001-01-01T00:00:00	0.936888	0.007833
1	5.849344	2020-11-25T13:00:00	0.997406	-0.023046
2	16.578772	0001-01-01T00:00:00	0.787193	-0.000694
3	7.082947	0001-01-01T00:00:00	0.733812	0.060157
4	6.341054	0001-01-01T00:00:00	0.311122	-0.010301

2.1 Rename columns to English

Use only the actual API fields. Filling values remain fractions: **0.60 means 60%**. Their weekly change is a difference in fractions, not measured water inflow. TWh describes energy storage potential, not water volume.

In [3]:
```
names = {
    "dato_Id": "date_id",
    "omrType": "area_type",
    "omrnr": "area_number",
    "iso_aar": "iso_year",
    "iso_uke": "iso_week",
    "fyllingsgrad": "fill_ratio",
    "kapasitet_TWh": "capacity_twh",
    "fylling_TWh": "storage_twh",
    "neste_Publiseringsdato": "next_publication_date",
    "fyllingsgrad_forrige_uke": "fill_ratio_previous_week",
    "endring_fyllingsgrad": "fill_ratio_change",
}
reservoirs = raw.rename(columns=names).copy()
reservoirs["date_id"] = pd.to_datetime(reservoirs["date_id"])
# The API uses year 0001 when a publication date is unavailable.
reservoirs["next_publication_date"] = pd.to_datetime(
    reservoirs["next_publication_date"], errors="coerce")
reservoirs = reservoirs.sort_values(["area_type", "area_number", "date_id"])
display(pd.Series(names, name="English name").rename_axis("API
field").to_frame())
```

```
/tmp/ipykernel_929/1562076016.py:17: UserWarning: Could not infer format, so
each element will be parsed individually, falling back to `dateutil`. To
ensure parsing is consistent and as-expected, please specify a format.
  reservoirs["next_publication_date"] = pd.to_datetime(
```

API field	English name
dato_Id	date_id
omrType	area_type
omrnr	area_number
iso_aar	iso_year
iso_uke	iso_week
fyllingsgrad	fill_ratio
kapasitet_TWh	capacity_twh
fylling_TWh	storage_twh
neste_Publiseringsdato	next_publication_date
fyllingsgrad_forrige_uke	fill_ratio_previous_week
endring_fyllingsgrad	fill_ratio_change

2.2 Inspect the data

Print a preview, field types, available area numbers and summary statistics. Retain the API observation date; ISO year/week are supporting metadata.

In [4]:
```
reservoirs.head()
reservoirs.dtypes.rename("Type").to_frame()
reservoirs.groupby("area_type")["area_number"].unique().to_frame()
print("Date range:", reservoirs.date_id.min(), "to",
reservoirs.date_id.max())
reservoirs.describe()
```

```
Date range: 1995-01-08 00:00:00 to 2026-10-11 00:00:00
```

Out[4]: Columns 1–8 of 11

	date_id	area_number	iso_year	iso_week	fill_ratio	capacity_twh	storage_twh
count	14922	14922.000000	14922.000000	14922.000000	14922.000000	14922.000000	14922.000000
mean	2010-11-24 12:00:00	2.333333	2010.393245	26.443908	0.618948	29.145860	18.142297
min	1995-01-08 00:00:00	0.000000	1995.000000	1.000000	0.051653	6.003264	0.310089
25%	2002-12-15 00:00:00	1.000000	2002.000000	13.000000	0.456817	17.390587	7.325947
50%	2010-11-24 12:00:00	2.000000	2010.000000	26.000000	0.648094	23.237715	14.504506
75%	2018-11-04 00:00:00	3.000000	2018.000000	39.000000	0.801885	34.043590	22.203242
max	2026-10-11 00:00:00	5.000000	2026.000000	53.000000	1.020072	87.437580	84.518060
std	NaN	1.490762	9.173142	15.011341	0.214686	22.739068	15.962980

Columns 9–11 of 11 · first column repeated

	next_publication_date	fill_ratio_previous_week	fill_ratio_change
count	3600	14922.000000	14922.000000
mean	2022-12-25 01:21:36	0.618940	0.000007
min	2019-02-27 13:00:00	0.051653	-0.242484
25%	2021-01-25 19:00:00	0.456817	-0.022360
50%	2022-12-25 01:00:00	0.648094	-0.006960
75%	2024-11-22 07:00:00	0.801785	0.016274
max	2026-10-21 13:00:00	1.020037	0.345632
std	NaN	0.214672	0.031334

3. NVE: plot each column appropriately

Choose one area for time-series plots. The default `NO`, area `0`, is the national series. Set `area_type = "EL"` and an available area number to study a price area. Do not join different areas into one line.

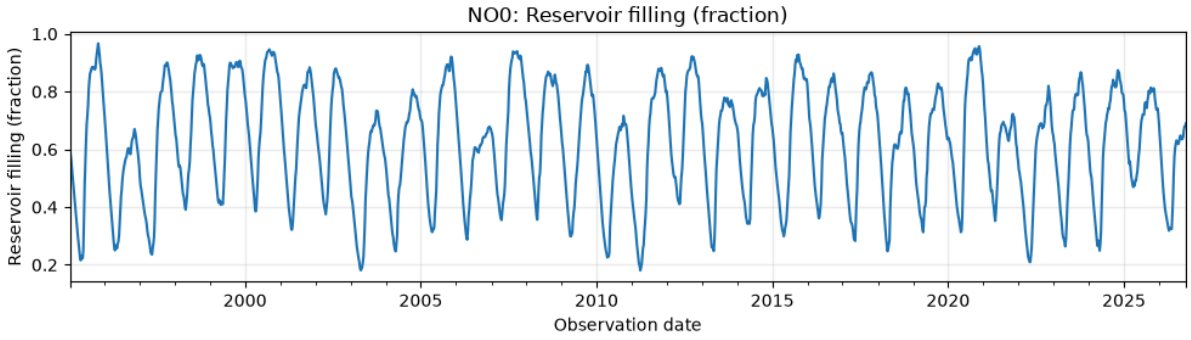
In [5]:
```
area_type = "NO"
area_number = 0
series = reservoirs.loc[
    reservoirs.area_type.eq(area_type) &
reservoirs.area_number.eq(area_number)
].sort_values("date_id").set_index("date_id")
assert not series.empty, "Choose an available area type and number."

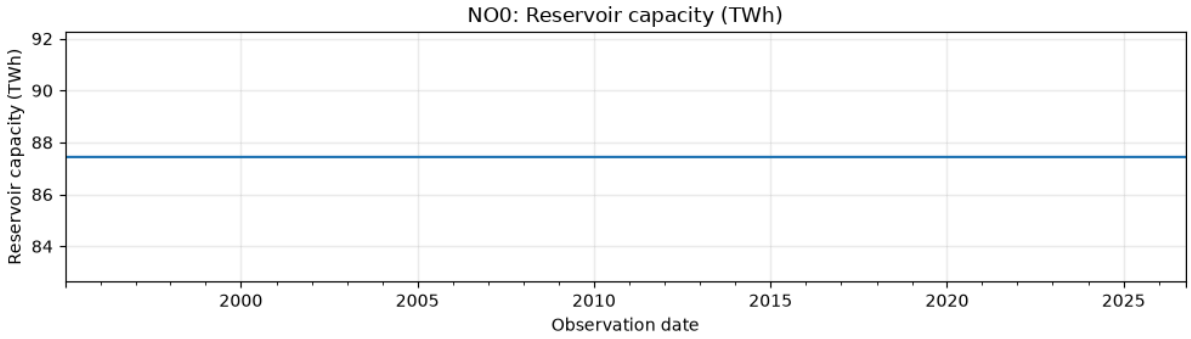
# Choose only those meaningful
metrics = {
    "fill_ratio": "Reservoir filling (fraction)",
    "capacity_twh": "Reservoir capacity (TWh)",
    "storage_twh": "Stored energy (TWh)",
    "fill_ratio_previous_week": "Previous week filling (fraction)",
    "fill_ratio_change": "Weekly filling change (fraction difference)",
}
series[list(metrics)].describe()
```

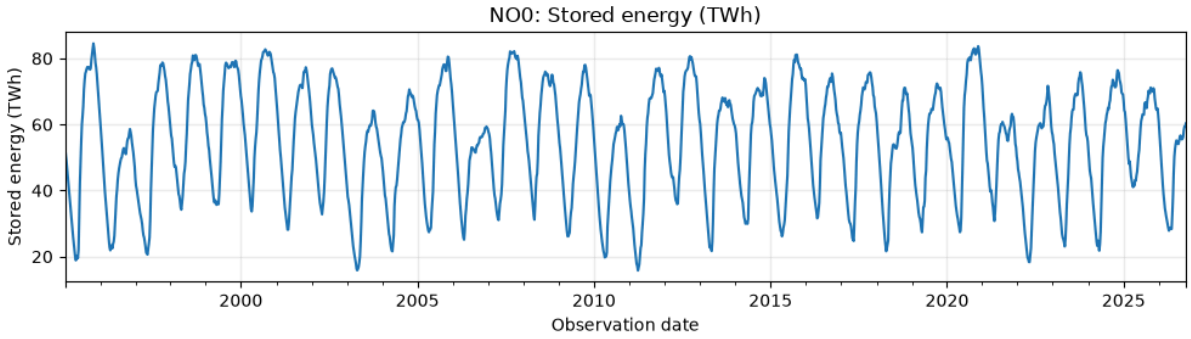
Out[5]:

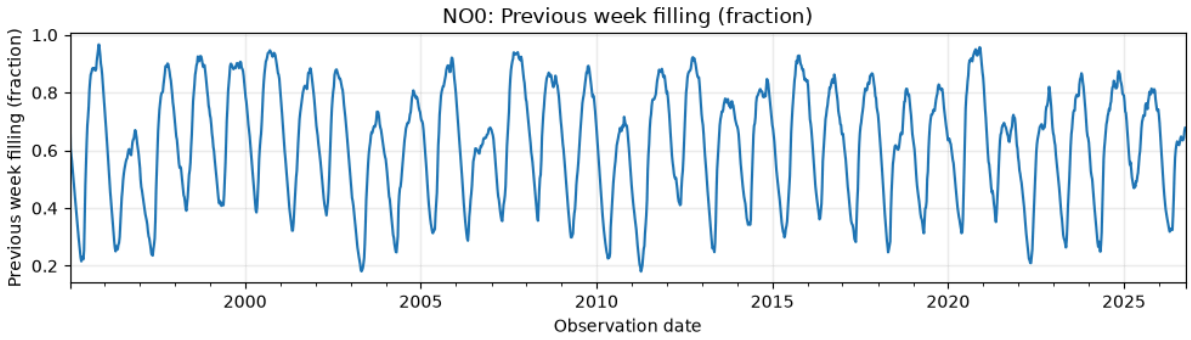
	fill_ratio	capacity_twh	storage_twh	fill_ratio_previous_week	fill_ratio_change
count	1658.000000	1.658000e+03	1658.000000	1658.000000	1658.000000
mean	0.622169	8.743758e+01	54.400922	0.622206	-0.000037
std	0.195691	1.421514e-14	17.110788	0.195710	0.027673
min	0.180250	8.743758e+01	15.760602	0.180260	-0.144643
25%	0.461099	8.743758e+01	40.317419	0.461099	-0.022020
50%	0.647552	8.743758e+01	56.620398	0.647567	-0.005516
75%	0.788700	8.743758e+01	68.962040	0.788699	0.016301
max	0.966610	8.743758e+01	84.518060	0.966610	0.125774

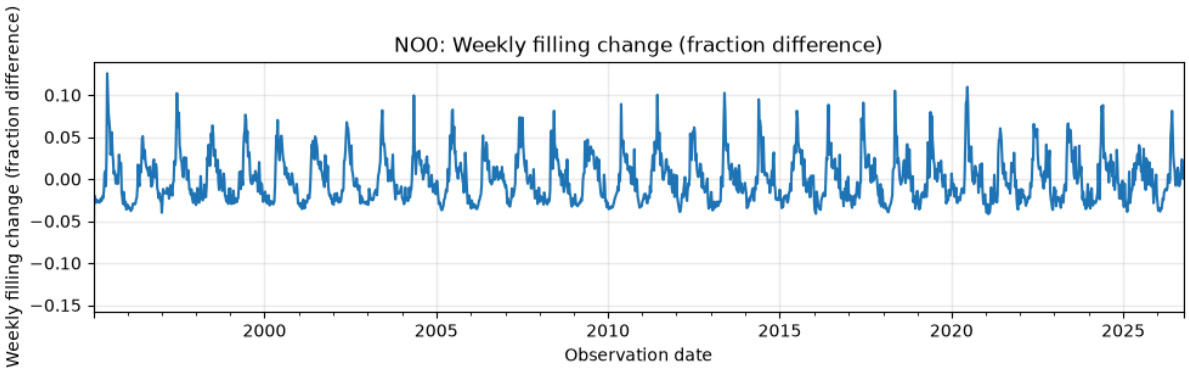
In [6]:
```
for column, label in metrics.items():
    ax = series[column].plot(figsize=(10, 3), title=f"{area_type}
{area_number}: {label}")
    ax.set(xlabel="Observation date", ylabel=label)
    ax.grid(alpha=0.25)
    plt.tight_layout()
    plt.show()
    plt.close()
```











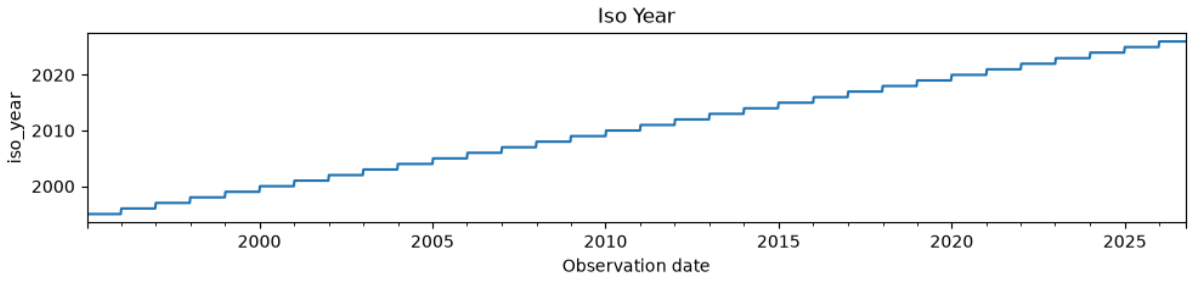
3.1 Dates, identifiers and calendar columns

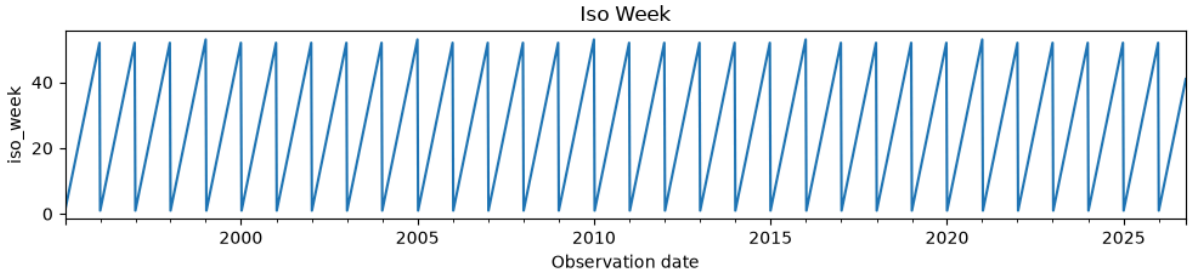
These columns describe observations rather than physical reservoir measurements. Plot counts for categories, calendar metadata against the observation date, and publication delay where a valid publication date exists. Area numbers are interpreted together with their area type.

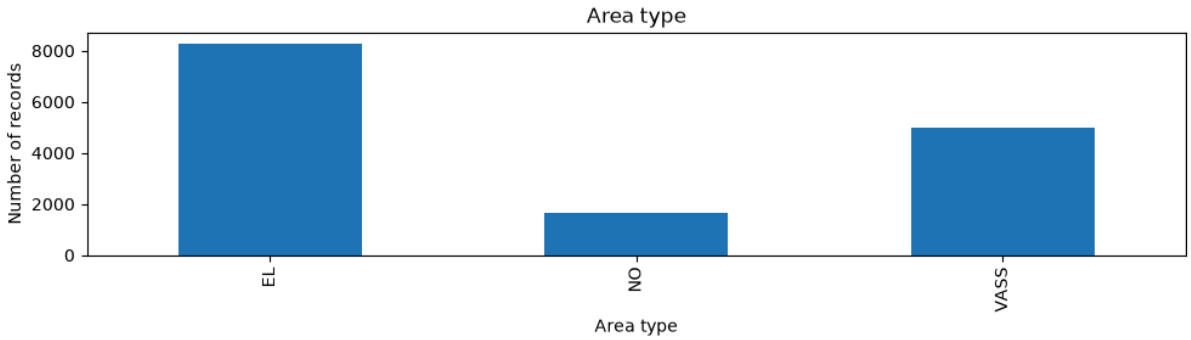
In [7]:
```
for column in ["iso_year", "iso_week"]:
    ax = series[column].plot(figsize=(10, 2.5), title=column.replace("_", "
").title())
    ax.set(xlabel="Observation date", ylabel=column)
    plt.tight_layout()
    plt.show()
    plt.close()

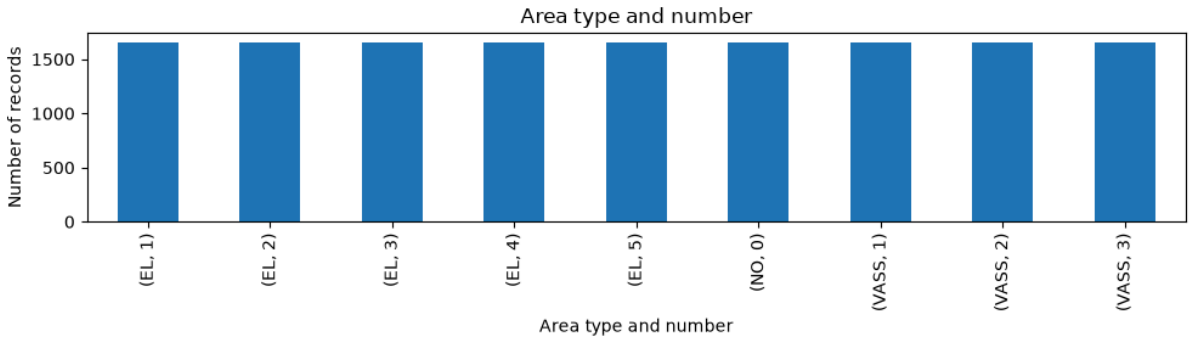
counts = {
    "Area type": reservoirs.area_type.value_counts().sort_index(),
    "Area type and number": reservoirs.groupby(["area_type",
"area_number"]).size(),
    "Observation date: records per year":
reservoirs.groupby(reservoirs.date_id.dt.year).size(),
}
for label, values in counts.items():
    ax = values.plot.bar(figsize=(10, 3), title=label)
    ax.set(xlabel=label, ylabel="Number of records")
    plt.tight_layout()
    plt.show()
    plt.close()

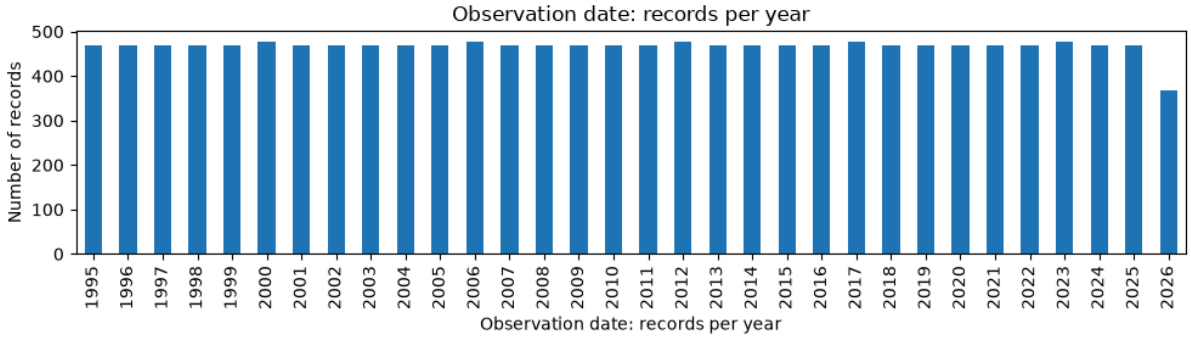
delay = (series.next_publication_date - series.index).dt.total_seconds() /
86400
ax = delay.dropna().plot(figsize=(10, 3), title="Next publication date
relative to observation")
ax.set(xlabel="Observation date", ylabel="Days until next publication")
plt.tight_layout()
plt.show()
plt.close()
```

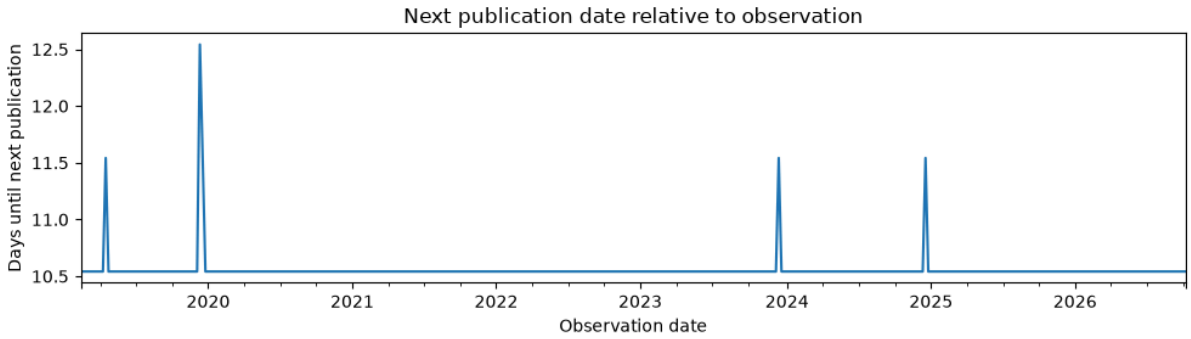








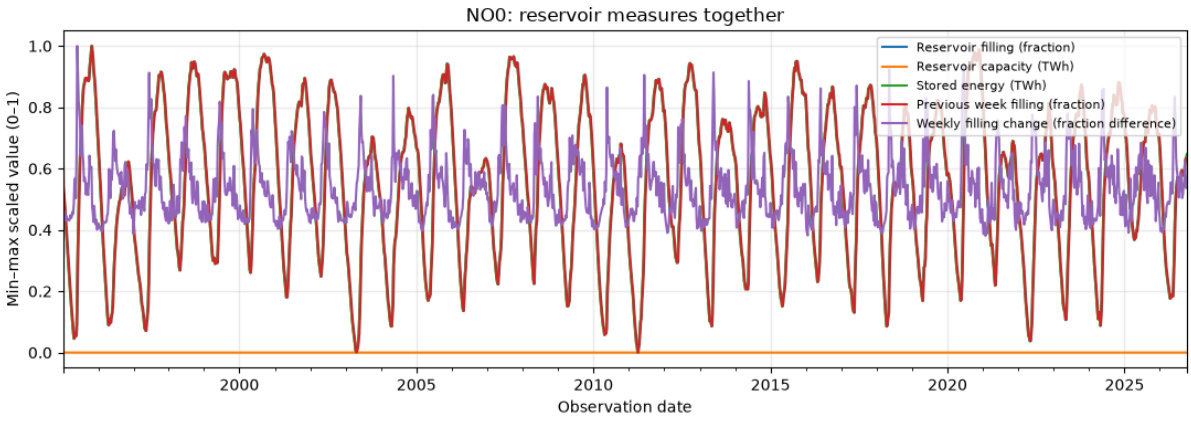




3.2 Plot all reservoir measures together

Capacity, stored energy and filling fractions have different scales. Min–max scaling places each physical measure between 0 and 1 over this selected series. A constant measure is shown at zero. Calendar fields and identifiers are excluded because they are not reservoir measurements. The chart compares patterns, not absolute magnitudes.

In [8]:
```
values = series[list(metrics)]
spread = values.max() - values.min()
scaled = (values - values.min()) / spread.replace(0, 1)
ax = scaled.rename(columns=metrics).plot(figsize=(11, 4))
ax.set(title=f"{area_type}{area_number}: reservoir measures together",
       xlabel="Observation date", ylabel="Min-max scaled value (0-1)")
ax.legend(fontsize=8)
ax.grid(alpha=0.25)
plt.tight_layout()
plt.show()
plt.close()
```



4. ENTSO-E: fetch cross-border transfers

ENTSO-E → Spark → Cassandra → Spark → MongoDB → Streamlit

Keep one connection for a simple example: **Norway NO2 ↔ Denmark DK1**. `out` means leaving Norway; `in` means entering Norway. This is not a complete national import/export dataset. The period is the last two years, ending at the start of today in Europe/Oslo. Enter credentials privately when prompted.

In [9]:
```
import os
import sys
import xml.etree.ElementTree as ET
from getpass import getpass
from urllib.parse import quote_plus
import pandas as pd
import matplotlib.pyplot as plt
from entsoe import EntsoeRawClient
from cassandra.cluster import Cluster
from pyspark.sql import SparkSession, functions as F
from pymongo import MongoClient, UpdateOne

token = getpass("ENTSO-E token: ")
password = quote_plus(getpass("Atlas database password: "))
uri = f"mongodb+srv://lovisanguyen_db_user:
{password}@cluster0.xvxeeyn.mongodb.net/?authSource=admin"
start = pd.Timestamp.now(tz="Europe/Oslo").normalize() -
pd.DateOffset(years=2)
end = pd.Timestamp.now(tz="Europe/Oslo").normalize()
```

```
ENTSO-E token:  ·····
Atlas database password:  ·····
```

4.1 Read the reported intervals

The helper retains interval duration, including quarter-hour observations and A03 variable blocks. Power is MW; **energy (MWh) = power (MW) × duration (hours)**. Missing quantities are not replaced with zero. Source revisions and unavailable periods should be reviewed before interpreting annual totals.

In [10]:
```
def read_intervals(xml, direction, left, right):
    rows = []
    for series in ET.fromstring(xml).findall("{*}TimeSeries"):
        assert series.findtext("{*}quantity_Measure_Unit.name") == "MAW",
"Expected MW data"
        curve = series.findtext("{*}curveType", "A01")
        assert curve in ("A01", "A03"), "Unsupported curve type"
        for period in series.findall("{*}Period"):
            begin = pd.Timestamp(period.findtext("
{*}timeInterval/{*}start"))
            finish = pd.Timestamp(period.findtext("{*}timeInterval/{*}end"))
            step = pd.Timedelta(period.findtext("{*}resolution"))
            points = sorted((int(p.findtext("{*}position")), p.findtext("
{*}quantity"))
                            for p in period.findall("{*}Point"))
            for i, (position, value) in enumerate(points):
                a = begin + (position - 1) * step
                b = min(a + step, finish)
                if curve == "A03":
                    b = begin + (points[i+1][0] - 1) * step if i+1 <
len(points) else finish
                a, b = max(a, left), min(b, right)
                if value is not None and a < b:
                    rows.append(("NO2-DK1", direction, a.isoformat(),
                                 float(value), (b-a).total_seconds()/3600))
    return rows
```

In [11]:
```
client = EntsoeRawClient(api_key=token)
rows = []
left = start
while left < end:
    right = min(left.replace(day=1) + pd.DateOffset(months=1), end)
    for direction, source, target in [("out", "NO_2", "DK_1"), ("in",
"DK_1", "NO_2")]:
        xml = client.query_crossborder_flows(source, target, start=left,
end=right)
        rows += read_intervals(xml, direction, left, right)
    left = right
print("Fetched intervals:", len(rows))

assert rows, "No intervals returned for the selected connection and period."
```

```
Fetched intervals: 60039
```

5. Spark: prepare and store in Cassandra

Spark parses UTC timestamps, assigns the local calendar year and calculates energy. Cassandra runs on your local machine; Streamlit Community Cloud does not provide this server.

> Before running the 'jupyter lab' comment in Ubuntu Terminal, remember to call out the 'conda activate nve-spark' which enable the environment from the yaml-file Or else

> PySparkRuntimeError: [JAVA_GATEWAY_EXITED] Java gateway process exited before sending its port number.

In [12]:
```
os.environ["PYSPARK_PYTHON"] = sys.executable
spark = (SparkSession.builder.master("local[2]").appName("SimpleENTSOE")
    .config("spark.jars.packages", "com.datastax.spark:spark-cassandra-
connector_2.12:3.5.1")
    .config("spark.cassandra.connection.host", "127.0.0.1")
    .config("spark.cassandra.connection.port", "9042")
    .config("spark.sql.session.timeZone", "UTC")
    .getOrCreate())

data = spark.createDataFrame(rows, "connection string, direction string,
time string, power_mw double, hours double")
data = (data.withColumn("time", F.to_timestamp("time"))
    .withColumn("year", F.year(F.from_utc_timestamp("time", "Europe/Oslo")))
    .withColumn("energy_mwh", F.col("power_mw") * F.col("hours"))
    .dropDuplicates(["connection", "direction", "time"]))
data.show(5)
```

```
26/10/19 11:22:18 WARN Utils: Your hostname, OSAI resolves to a loopback
address: 127.0.1.1; using 10.255.255.254 instead (on interface lo)
26/10/19 11:22:18 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to
another address
```

```
:: loading settings :: url = jar:file:/home/tyyniuni/miniconda3/envs/nve-
spark/lib/python3.11/site-packages/pyspark/jars/ivy-
2.5.3.jar!/org/apache/ivy/core/settings/ivysettings.xml
```

```
Ivy Default Cache set to: /home/tyyniuni/.ivy2/cache
The jars for the packages stored in: /home/tyyniuni/.ivy2/jars
com.datastax.spark#spark-cassandra-connector_2.12 added as a dependency
:: resolving dependencies :: org.apache.spark#spark-submit-parent-142983ee-
89fb-4f8e-952a-8bbd9128f9d1;1.0
        confs: [default]
        found com.datastax.spark#spark-cassandra-connector_2.12;3.5.1 in
central
        found com.datastax.spark#spark-cassandra-connector-driver_2.12;3.5.1
in central
        found org.scala-lang.modules#scala-collection-compat_2.12;2.11.0 in
central
        found org.apache.cassandra#java-driver-core-shaded;4.18.1 in central
        found com.datastax.oss#native-protocol;1.5.1 in central
        found com.datastax.oss#java-driver-shaded-guava;25.1-jre-graal-sub-1
in central
        found com.typesafe#config;1.4.1 in central
        found org.slf4j#slf4j-api;1.7.26 in central
        found io.dropwizard.metrics#metrics-core;4.1.18 in central
        found org.hdrhistogram#HdrHistogram;2.1.12 in central
        found org.reactivestreams#reactive-streams;1.0.3 in central
        found org.apache.cassandra#java-driver-mapper-runtime;4.18.1 in
central
        found org.apache.cassandra#java-driver-query-builder;4.18.1 in
central
        found org.apache.commons#commons-lang3;3.10 in central
        found com.thoughtworks.paranamer#paranamer;2.8 in central
```

```
        found org.scala-lang#scala-reflect;2.12.19 in central
:: resolution report :: resolve 1376ms :: artifacts dl 58ms
        :: modules in use:
        com.datastax.oss#java-driver-shaded-guava;25.1-jre-graal-sub-1 from
central in [default]
        com.datastax.oss#native-protocol;1.5.1 from central in [default]
        com.datastax.spark#spark-cassandra-connector-driver_2.12;3.5.1 from
central in [default]
        com.datastax.spark#spark-cassandra-connector_2.12;3.5.1 from central
in [default]
        com.thoughtworks.paranamer#paranamer;2.8 from central in [default]
        com.typesafe#config;1.4.1 from central in [default]
        io.dropwizard.metrics#metrics-core;4.1.18 from central in [default]
        org.apache.cassandra#java-driver-core-shaded;4.18.1 from central in
[default]
        org.apache.cassandra#java-driver-mapper-runtime;4.18.1 from central
in [default]
        org.apache.cassandra#java-driver-query-builder;4.18.1 from central
in [default]
        org.apache.commons#commons-lang3;3.10 from central in [default]
        org.hdrhistogram#HdrHistogram;2.1.12 from central in [default]
        org.reactivestreams#reactive-streams;1.0.3 from central in [default]
        org.scala-lang#scala-reflect;2.12.19 from central in [default]
        org.scala-lang.modules#scala-collection-compat_2.12;2.11.0 from
central in [default]
        org.slf4j#slf4j-api;1.7.26 from central in [default]
        ---------------------------------------------------------------------
-
        |                  |            modules            ||   artifacts
|
        |       conf       | number| search|dwnlded|evicted||
number|dwnlded|
        ---------------------------------------------------------------------
-
        |      default     |   16  |   0   |   0   |   0   ||   16  |   0
|
        ---------------------------------------------------------------------
-
:: retrieving :: org.apache.spark#spark-submit-parent-142983ee-89fb-4f8e-
952a-8bbd9128f9d1
        confs: [default]
        0 artifacts copied, 16 already retrieved (0kB/15ms)
26/10/19 11:22:21 WARN NativeCodeLoader: Unable to load native-hadoop
library for your platform... using builtin-java classes where applicable
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use
setLogLevel(newLevel).
26/10/19 11:22:36 WARN TaskSetManager: Stage 0 contains a task of very large
size (1572 KiB). The maximum recommended task size is 1000 KiB.
```

```
+----------+---------+-------------------+--------+-----+----+----------+
|connection|direction|               time|power_mw|hours|year|energy_mwh|
+----------+---------+-------------------+--------+-----+----+----------+
```

```
|   NO2-DK1|      out|2024-10-22 09:00:00| 1622.94|  1.0|2024|   1622.94|
|   NO2-DK1|      out|2024-10-22 23:00:00|  1632.0|  1.0|2024|    1632.0|
|   NO2-DK1|       in|2024-12-17 02:00:00|  487.85|  1.0|2024|    487.85|
|   NO2-DK1|       in|2025-01-13 17:00:00|  253.36|  1.0|2025|    253.36|
|   NO2-DK1|      out|2025-04-09 15:30:00|  699.75| 0.25|2025|  174.9375|
+----------+---------+-------------------+--------+-----+----+----------+
only showing top 5 rows
```

5.1 Write using Spark

Use a separate table for each notebook run to avoid mixing current observations with old results. The UUID produces a valid table suffix. Old tables remain available for inspection; no existing data is deleted.

> Remember to turn the Docker Image of Cassandra before, or else the error
>
> > IF error, run the import socket cell one time to re-awake the ports, and then the cluster cell.

In [15]:
```python
import socket

with socket.create_connection(("127.0.0.1", 9042), timeout=10):
    print("Cassandra port is reachable")
```

```
Cassandra port is reachable
```

In [16]:
```
from cassandra.cluster import Cluster
from cassandra.policies import RoundRobinPolicy

cluster = Cluster(
    ["127.0.0.1"],
    port=9042,
    protocol_version=5,
    load_balancing_policy=RoundRobinPolicy()
)
session = cluster.connect()

session.execute("""CREATE KEYSPACE IF NOT EXISTS nve
    WITH replication = {'class':'SimpleStrategy',
'replication_factor':1}""")
from uuid import uuid4
table = "simple_transfers_" + uuid4().hex
session.execute(f"""CREATE TABLE IF NOT EXISTS nve.{table} (
    connection text, direction text, year int, time timestamp,
    power_mw double, hours double, energy_mwh double,
    PRIMARY KEY ((connection, direction, year), time)
)""")
(data.write.format("org.apache.spark.sql.cassandra")
    .options(keyspace="nve", table=table).mode("append").save())
```

```
/tmp/ipykernel_929/4236473552.py:4: DeprecationWarning: Legacy execution
parameters will be removed in 4.0. Consider using execution profiles.
  cluster = Cluster(
26/10/19 11:24:39 WARN TaskSetManager: Stage 3 contains a task of very large
size (1572 KiB). The maximum recommended task size is 1000 KiB.
```

5.2 Read Cassandra and extract relevant columns using Spark

The chart and MongoDB input must come from this readback, not directly from the API response. Compare the readback with the prepared data before continuing.

In [17]:
```
extracted = (
    spark.read.format("org.apache.spark.sql.cassandra")
    .options(keyspace="nve", table=table).load()
    .filter(F.col("connection") == "NO2-DK1")
    .filter(F.col("direction").isin("in", "out"))
    .filter(F.col("year").isin(
        list(range(start.year, (end - pd.Timedelta(seconds=1)).year + 1))
    ))
    .filter(
        (F.col("time") >= F.lit(start.to_pydatetime()))
        & (F.col("time") < F.lit(end.to_pydatetime()))
    )
    .select(
        "connection", "direction", "year", "time",
        "power_mw", "hours", "energy_mwh"
    )
)

# Preserve UTC when converting Spark timestamps to pandas.
df = extracted.withColumn("time", F.col("time").cast("long")).toPandas()
df["time"] = pd.to_datetime(df["time"], unit="s", utc=True)
df = df.sort_values("time")

df.head()
```

Out[17]:

	connection	direction	year	time	power_mw	hours	energy_mwh
0	NO2-DK1	in	2024	2024-10-18 22:00:00+00:00	0.00	34.0	0.00
18842	NO2-DK1	out	2024	2024-10-18 22:00:00+00:00	1385.20	1.0	1385.20
18843	NO2-DK1	out	2024	2024-10-18 23:00:00+00:00	1385.24	1.0	1385.24
18844	NO2-DK1	out	2024	2024-10-19 00:00:00+00:00	1385.07	1.0	1385.07
18845	NO2-DK1	out	2024	2024-10-19 01:00:00+00:00	1385.01	1.0	1385.01

6. Plot the Spark-extracted transfers

Choose the last completed year. The pie compares imports and exports as **energy**, not production. Report coverage alongside totals: missing intervals and incomplete years must not be presented as complete annual balances.

In [22]:

```
import matplotlib.pyplot as plt
import pandas as pd

# 1. Determine target year
year = end.year - 1
annual = df.loc[df.year.eq(year)]
assert not annual.empty, f"Choose a year with reported data (no data for
{year})."

# 2. Aggregate reported values
totals = annual.groupby("direction").agg(
    energy_mwh=("energy_mwh", "sum"),
    reported_hours=("hours", "sum")
)

# 3. Calculate expected calendar hours accurately (DST-neutral)
start_dt = pd.Timestamp(year=year, month=1, day=1)
end_dt = pd.Timestamp(year=year + 1, month=1, day=1)
expected_hours = (end_dt - start_dt).total_seconds() / 3600

# 4. Coverage calculation
totals["coverage_pct"] = totals.reported_hours / expected_hours * 100
display(totals)

if len(totals) < 2 or (totals.reported_hours < expected_hours).any():
    print("Incomplete coverage: totals represent reported observations
only.")

# 5. Plotting logic
if totals.energy_mwh.sum() > 0:
    # Rename categories safely
    series = totals.energy_mwh.rename(index={"in": "Into Norway", "out":
"Out of Norway"})

    # Map colors directly to categories to avoid positional color bugs
    color_map = {
        "Into Norway": "#005A9C",    # Deep Blue
        "Out of Norway": "#E05A47"   # Warm Red/Coral
    }
    plot_colors = [color_map.get(label, "#95A5A6") for label in
series.index]

    fig, ax = plt.subplots(figsize=(7, 6), dpi=300, facecolor="#FCFCFC")
    ax.set_facecolor("#FCFCFC")

    wedges, texts, autotexts = ax.pie(
        series,
        labels=series.index,
        autopct="%1.1f%%",
        startangle=140,
        colors=plot_colors,
        pctdistance=0.75,
```

```
        wedgeprops={
            "width": 0.45,
            "edgecolor": "white",
            "linewidth": 2.5,
            "antialiased": True,
        },
        textprops={"fontsize": 11, "fontweight": "medium", "color":
"#2C3E50"},
    )

    for autotext in autotexts:
        autotext.set_color("white")
        autotext.set_fontsize(11)
        autotext.set_weight("bold")

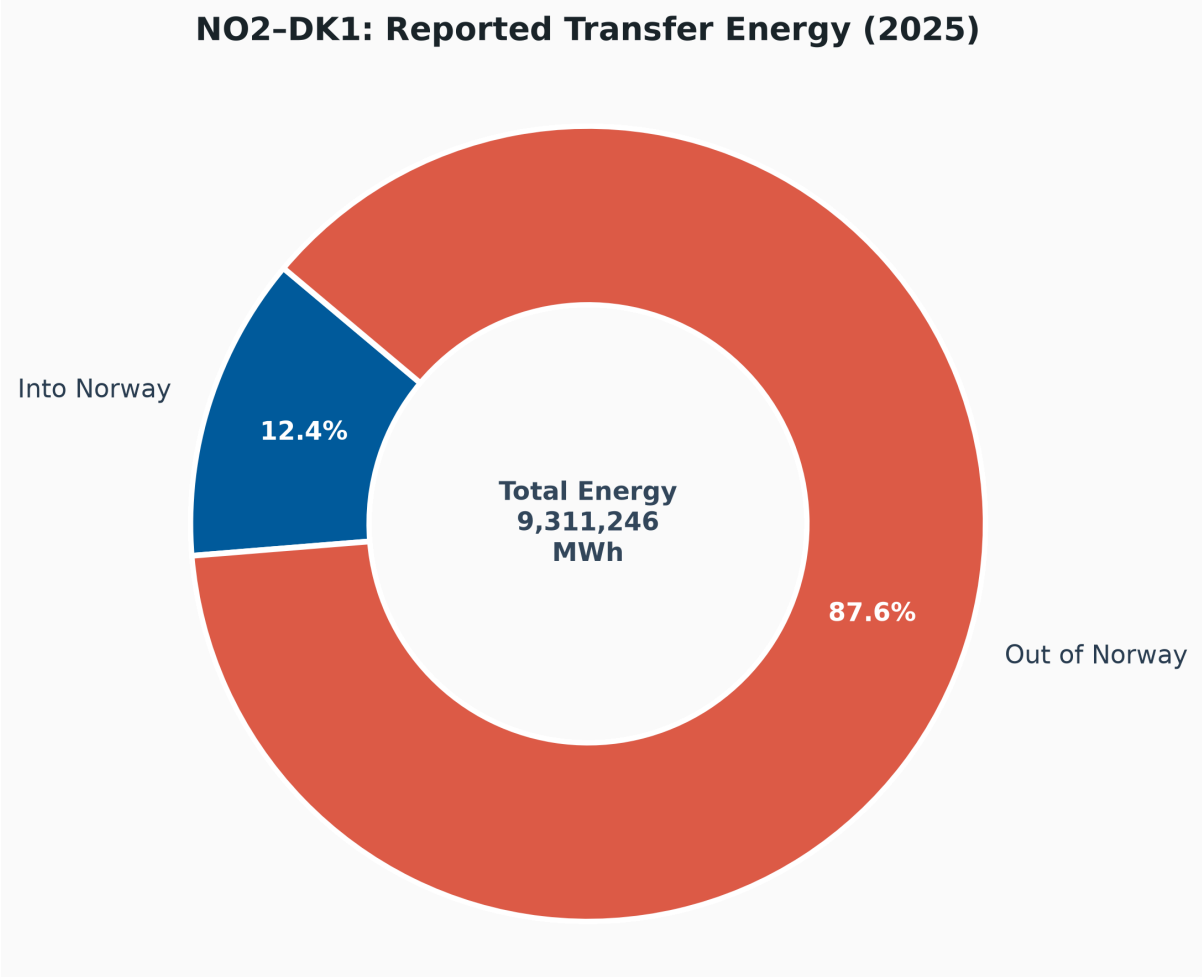
    ax.set_title(
        f"NO2-DK1: Reported Transfer Energy ({year})",
        fontsize=14,
        fontweight="bold",
        color="#1A252C",
        pad=20,
        loc="center",
    )

    total_val = series.sum()
    ax.text(
        0,
        0,
        f"Total Energy\n{total_val:,.0f}\nMWh",
        ha="center",
        va="center",
        fontsize=11,
        fontweight="bold",
        color="#34495E",
    )

    ax.set_ylabel("")
    ax.axis("equal")

    plt.tight_layout()
    plt.show()
    plt.close()
```

	energy_mwh	reported_hours	coverage_pct
direction			
in	1.153915e+06	8760.0	100.0
out	8.157331e+06	8760.0	100.0



6.1 January: one line for each direction

Draw each reported interval at its measured power. NaN separators prevent invented lines across missing intervals. The calendar month uses Europe/Oslo; the plotted timestamps remain UTC.

In [23]:

```
# 1. Ensure time is datetime and converted to local time (Europe/Oslo)
annual_local = annual.copy()
annual_local["time_local"] =
annual_local["time"].dt.tz_convert("Europe/Oslo")

# 2. Filter January in local time
january =
annual_local.loc[annual_local["time_local"].dt.month.eq(1)].sort_values("tim
e_local")

# Direction label mapping & color scheme
direction_labels = {"in": "Into Norway (Import)", "out": "Out of Norway
(Export)"}
colors = {"in": "#005A9C", "out": "#E05A47"}

fig, ax = plt.subplots(figsize=(12, 4.5), dpi=300, facecolor="#FCFCFC")
ax.set_facecolor("#FCFCFC")

# 3. Vectorized step plot using drawstyle='steps-post'
```

```
for direction, group in january.groupby("direction"):
    label_text = direction_labels.get(direction, direction)
    color = colors.get(direction, None)

    ax.plot(
        group["time_local"],
        group["power_mw"],
        drawstyle="steps-post",
        label=f"NO2-DK1: {label_text}",
        linewidth=1.2,
        color=color,
        alpha=0.85
    )

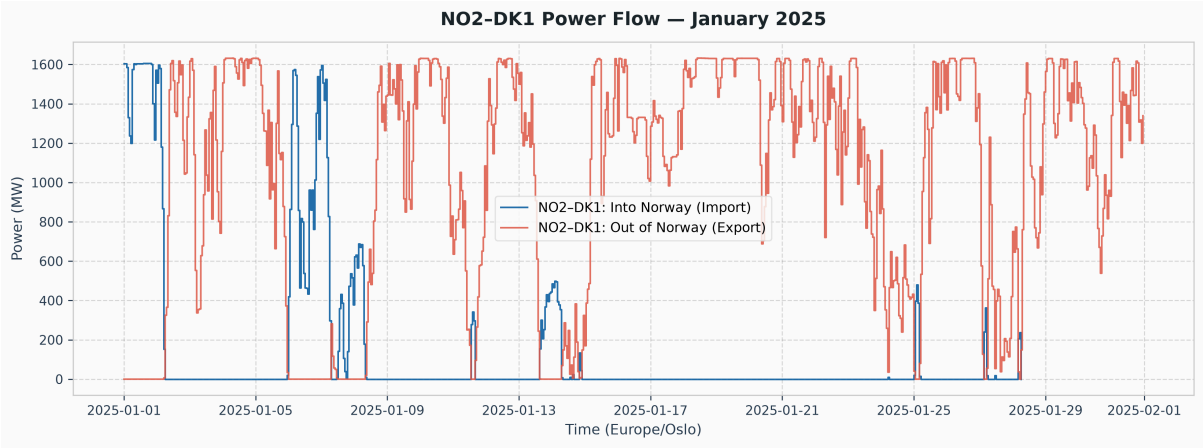
# 4. Clean labels and formatting
ax.set_title(f"NO2-DK1 Power Flow — January {year}", fontsize=13,
fontweight="bold", pad=12, color="#1A252C")
ax.set_xlabel("Time (Europe/Oslo)", fontsize=10, fontweight="medium",
color="#2C3E50")
ax.set_ylabel("Power (MW)", fontsize=10, fontweight="medium",
color="#2C3E50")

ax.legend(frameon=True, facecolor="white", edgecolor="#E0E0E0",
fontsize=9.5)
ax.grid(True, linestyle="--", alpha=0.3, color="#888888")

# Style tick labels
ax.tick_params(colors="#2C3E50", labelsize=9)
for spine in ax.spines.values():
    spine.set_color("#CCCCCC")

plt.tight_layout()
plt.show()
plt.close()
```

```
findfont: Failed to find font weight medium for DejaVu Sans, now using 400.
```



6.2 Save the Spark-extracted data to MongoDB

Upserts make repeated publication safe for the same connection, direction and timestamp. The app filters the current requested window; records from earlier periods may remain in the collection. Publication uses batches and is not an atomic snapshot: publish with the dashboard idle and confirm the saved row count before sharing results.

In [20]:
```
mongo = MongoClient(uri)
mongo.admin.command("ping")
collection = mongo["nve"]["simple_transfers"]
collection.create_index([("connection",1), ("direction",1), ("time",1)],
unique=True)
documents = df.to_dict("records")
for offset in range(0, len(documents), 1000):
    batch = documents[offset:offset+1000]
    collection.bulk_write([UpdateOne(
        {"connection":d["connection"], "direction":d["direction"],
"time":d["time"]},
        {"$set":d}, upsert=True) for d in batch])
print("Saved records:", len(documents))
```

```
Saved records: 60039
```

In [21]:
```
saved = collection.count_documents({"connection":"NO2-DK1", "time":{
    "$gte":start.to_pydatetime(), "$lt":end.to_pydatetime()}})
print("MongoDB records in the requested period:", saved)
assert saved == len(df), "Check existing records and the published period."
```

```
MongoDB records in the requested period: 60039
```

7. Streamlit: four simple pages

The accompanying files implement the app tasks from `Dtb_ca1`:

Page	Contents
Home	Sidebar navigation and the two data flows
Reservoir table	First-month numeric sparklines; one row per imported field
Reservoir plots	Measure/all selector and monthly selection slider, default first month
Electricity transfers	MongoDB records, annual imports/exports pie and January direction lines

`load_reservoirs()` caches the NVE API response. The slider filters locally because this endpoint has no date parameters. One selected area is displayed at a time. All fields have table rows. Numeric metadata has labelled sparklines; dates and text use displayed values, not artificial numeric conversions.

In a terminal, run `streamlit run app.py` from the supplied folder. Copy `.streamlit/secrets.toml.example` to `.streamlit/secrets.toml` and enter your Atlas URI.

Streamlit reads MongoDB for transfers; it does not run Spark or Cassandra. Do not commit the real secrets file. No GitHub upload or deployment is performed by this notebook.

8. Work log

The original reservoir CSV is replaced by NVE's public API, while ENTSO-E supplies electricity-transfer observations. Then follows the guidances from compulsory assignment 1 & 2.

8.1 AI usage

Errors and Corrective Actions

Error or issue	Diagnosis and corrective action
`ModuleNotFoundError: typing.io`	Identified an incompatibility between the installed PySpark version and Python 3.14. Recommended a Python 3.11 environment.
Missing dependencies	Prepared pinned pip requirements. Clarified that Python 3.11 and Java 17 require separate installation.
`JAVA_GATEWAY_EXITED`	Investigated Java startup failure by checking the Java executable, `JAVA_HOME` and the notebook environment.
`UnsupportedOperationException: getSubject is not supported`	The log indicated a likely unsupported Java runtime. Provided code to select Java 17 explicitly before creating the Spark session.
`DATA_SOURCE_NOT_FOUND: org.apache.spark.sql.cassandra`	Identified that the active Spark session lacked the Cassandra connector. Recommended restarting the kernel and configuring the connector in the first Spark session.
`NoHostAvailable` / `OperationTimedOut`	Investigated connectivity to Cassandra at `127.0.0.1:9042`. Provided separate checks for container readiness and network access from Jupyter.
MongoDB connection configuration	Explained where to enter the Atlas URI, how to encode the database password, and how to configure Streamlit Secrets.
Undefined plotting variables	Revised code that referenced undefined variables, including `np` and `primary_date`.

Data Interpretation Improvements

Issue	Improvement
Different reservoir areas combined into one series	Selected one area at a time to avoid mixing overlapping classifications.
Measurements with different units plotted together	Used min–max scaling for comparisons and explained the loss of absolute magnitudes.
Reservoir storage described as water volume	Clarified that **TWh represents stored energy potential**, not physical water volume.

Issue	Improvement
Electricity transfers described as production	Distinguished imports and exports from domestic electricity generation.

Validation Status

Check	Status
Offline Streamlit checks	Passed using controlled test inputs
Local Spark transformations and energy calculations	Passed
Cassandra inside Docker	Responded with version **5.0.9**
Cassandra connectivity from Jupyter	Awaiting confirmation

Timezone & Hours Accounting (Bug Fix)

- **DST Shift Fix:** Replaced localized `Europe/Oslo` timestamp arithmetic with DST-neutral `pd.Timestamp` differences. Localized timezone math returned 8,759 or 8,761 hours due to spring/autumn clock shifts; standard UTC difference ensures exact 8,760 calendar hours (or 8,784 for leap years).
- **Labeling Consistency:** Standardized x-axis labels to match the timezone used during filtering (`Europe/Oslo`), avoiding misalignment near month boundaries.

Visualization & Color Logic (Accuracy & Aesthetics)

- **Explicit Category-Color Mapping:** Replaced positional color lists with dictionary mapping (`{"in": "#005A9C", "out": "#E05A47"}`). This prevents color flipping if a subset contains only one flow direction.
- **Modern Donut Architecture:** Transformed the default pie chart into a modern donut chart (`width=0.45`) featuring total energy (MWh) centered in the void, improving legibility and visual hierarchy.

Performance & Data Integrity

- **Vectorized Step Plots:** Replaced slow `itertuples()` loops with Matplotlib's native `drawstyle="steps-post"` , speeding up execution and rendering.
- **Temporal Sorting:** Enforced explicit `.sort_values("time")` prior to step rendering to eliminate erratic horizontal jump artifacts from unsorted rows.

8.2 References

- NVE Magasinstatistikk API
- ENTSO-E API reference
- Spark Cassandra connector
- Streamlit LineChartColumn

In [25]:
```
spark.stop()
cluster.shutdown()
mongo.close()
```